

## EFASOLAR PVStation Solutions



EFASOLAR PVStation Solutions

Efacec is a century-long experienced company in providing power conversion product solutions, we develop advanced fully integrated photovoltaic converter stations, covering the entire electric conversion chain for photovoltaic power plants.

EFASOLAR PVStation solution is highly competitive for large photovoltaic plants. A complete solution with in-house developed products, according to the highest efficiency and reliability standards, achieved through solar inverters, medium voltage switchgear, transformer, energy automation solutions and the enclosure.

With an entirely integrated approach, we are also able to provide photovoltaic grid-connection substations and interface solutions to the grid operator dispatch through its energy automation solutions.

Seeking for the highest performance during the EFASOLAR PVStation lifetime, optionally support services are available from base maintenance to a 24/7 protection approach and a multiple after sales packages assuring high availability and performance of our products.

| Rated Power                              | 1,00 MW                                                                | 2,00 MW       | 3,49 MW       | 4,00 MW        | 4,66 MW       |
|------------------------------------------|------------------------------------------------------------------------|---------------|---------------|----------------|---------------|
| EFASOLAR Inverter                        | 2 x 500M                                                               | 2 x 1000      | 3 x 1165      | 4 x 1000       | 4 x 1165      |
| Maximum PV power                         | 2 x 575 kW                                                             | 2 x 1200 kW   | 3 x 1400 kW   | 3 x 1200 kW    | 4 x 1400 kW   |
| MPPT Range                               | 480 V - 820 V                                                          | 625 V - 900 V | 630 V - 890 V | 625 V - 900 V  | 630 V - 890 V |
| Maximum DC current                       | 2 x 1084 A                                                             | 2 x 1600 A    | 3 x 1950 A    | 4 x 1600 A     | 4 x 1950 A    |
| Number of independent MPP inputs         | 4                                                                      | 2             | 3             | 4              | 4             |
| Number of DC fused inputs <sup>(1)</sup> | 16                                                                     | 12            | 18            | 24             | 24            |
| Transformer                              | Oil Immersed / Dry Type                                                |               |               |                |               |
| Power                                    | 1000 kVA                                                               | 2000 kVA      | 1 x 3490 kVA  | 2 x 2000 kVA   | 2 x 2330 kVA  |
| Number of LV Windings                    | 2                                                                      | 2             | 3             | 2              | 2             |
| Switchgear <sup>(2)</sup>                | 2x IS + CIS / IS + CIS                                                 |               |               | 2x IS + 2x CIS |               |
| Voltage                                  | ≤ 36 kV                                                                |               |               |                |               |
| Insulation                               | Air / SF6                                                              |               |               |                |               |
| Enclosure                                | Concrete / Container / Modular / Padmount UL                           |               |               |                |               |
| Dimensions (WxDxH)                       | 7500 x 2730 x 3200 mm / 20 ft -40 ft Container / 9000 x 2980 x 2950 mm |               |               |                |               |
| Weight                                   | ≤ 35 tons Concrete / ≤ 24 tons Container                               |               |               |                |               |
| Cooling                                  | Air forced cooling                                                     |               |               |                |               |
| Protection Degree                        | IP 23D                                                                 |               |               |                |               |

(1) - Other configurations can be used.  
(2) - Different configurations available upon request.

EFASOLAR PVStation

|         |                         |
|---------|-------------------------|
| 0,25 MW | 1 x EFASOLAR 250        |
| 0,50 MW | 1 x EFASOLAR 500M       |
| 1 MW    | 2 x EFASOLAR 500 / 500M |
| 1,46 MW | 2 x EFASOLAR 730        |
| 2 MW    | 2 x EFASOLAR 1000       |
| 2 MW    | 1 x EFASOLAR 2000       |
| 2,33 MW | 2 x EFASOLAR 1165       |
| 2,52 MW | 4 x EFASOLAR 630        |
| 3 MW    | 3 x EFASOLAR 1000       |
| 3,49 MW | 3 x EFASOLAR 1165       |
| 4 MW    | 4 x EFASOLAR 1000       |
| 4 MW    | 2 x EFASOLAR 2000       |
| 4,66 MW | 4 x EFASOLAR 1165       |

Efacec Inverter

- Multiple d.c. inputs & integrated fuses with individual current measurement
- Power control & grid support features

Efacec MV Transformer

- Oil
- Dry
- Outdoor
- Single Winding
- Double Winding

Efacec MV Switchgear ≤ 36 kV

Configurable Blocks

- Metering
- Compact
- Modular

After Sales

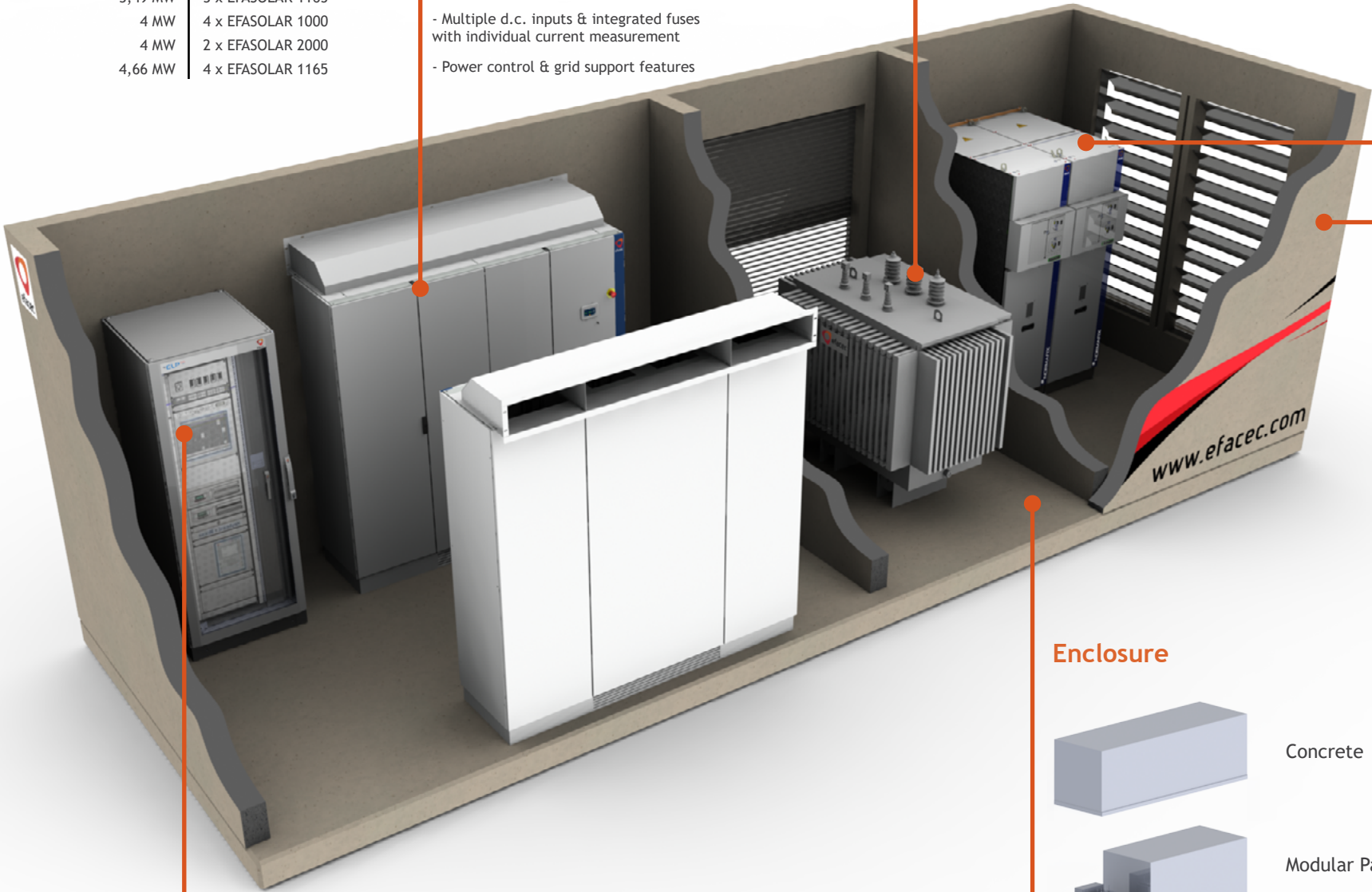
- Hotline
- Maintenance
- Life Cycle Warranty Options

Enclosure

- Concrete | Metal
- Modular Padmount UL
- Container 20ft. / 40ft.

Efacec SCADA Monitoring System

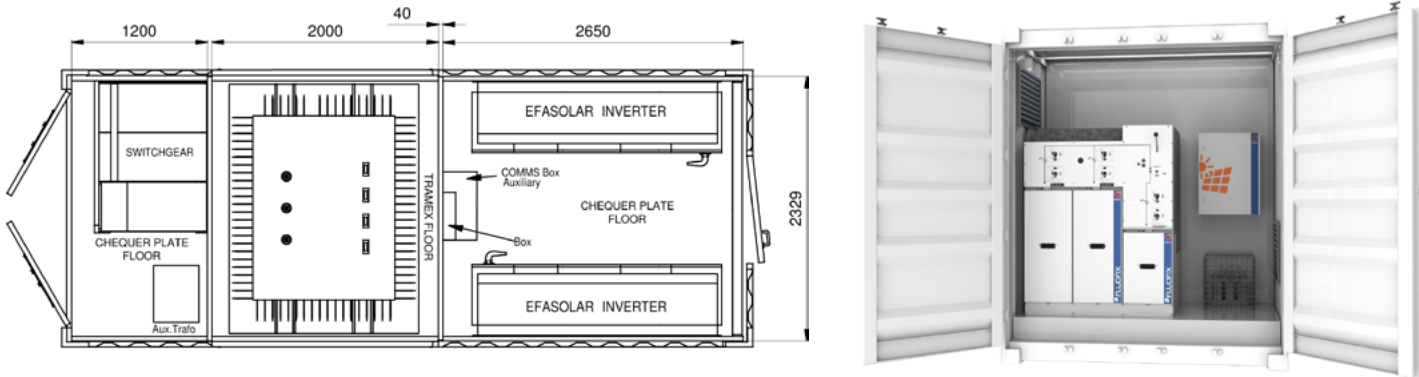
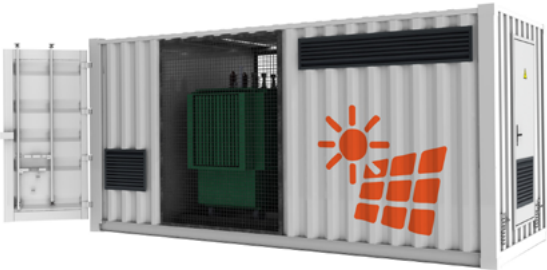
- Power Plant Controller
- String Monitoring Integration
- Dispach / DNO / ISO Interface
- High Reliability System (Hot-standby Redundancy)
- User Friendly Web based HMI



EFASOLAR PVStation 20ft Container

Efacec worldwide transportable and compact PVStation solution.

The best solution for large scale PV power plant where the transportation is a key point. With his standard container size is a plug & play solution for any kind of installation in any local conditions type.

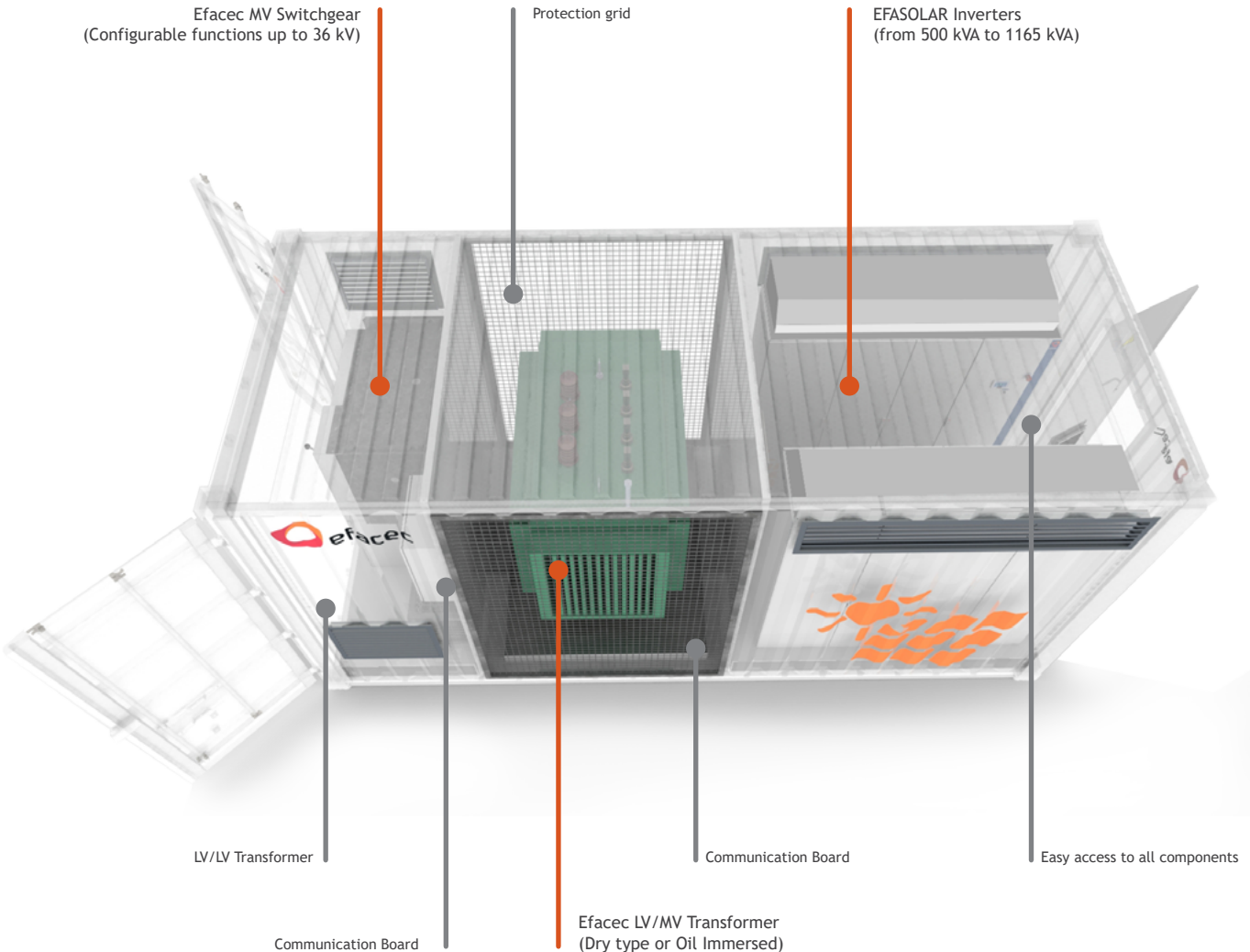


Customer Benefits

- 20ft solutions up to 2,33 MVA
- Reduced transportation and installation cost
- Plug & play solution
- Configurable MV switchgear functions
- High efficiency cooling system

Key Benefits

- All Efacec equipment are manufactured in-house
- Efacec single source of technology
- One single warranty for entire PVStation
- One service support for all products
- Robust and high quality container



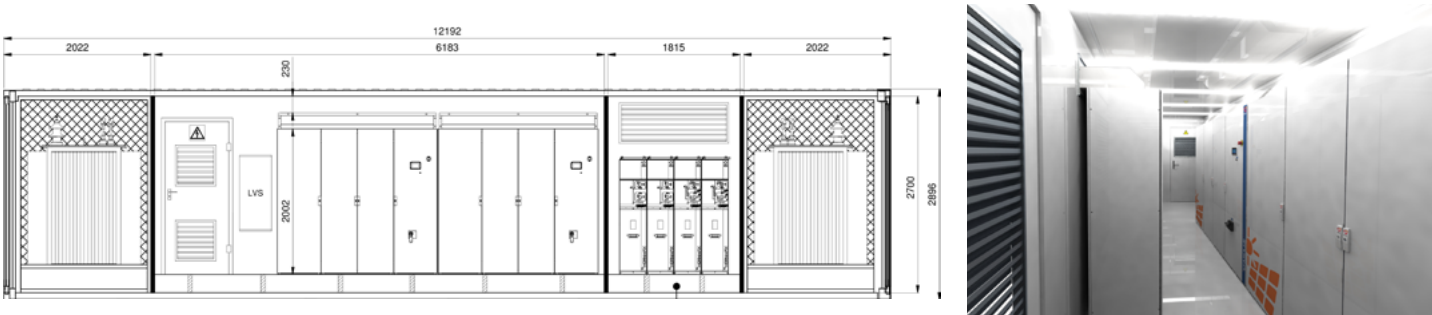
| Electrical                                                                                |                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC input                                                                                  |                                                                                                                                                                                                                                                                                 |
| Number of independent MPP inputs                                                          | 2                                                                                                                                                                                                                                                                               |
| Number of DC fused inputs <sup>(1)</sup>                                                  | up to 12                                                                                                                                                                                                                                                                        |
| AC output                                                                                 |                                                                                                                                                                                                                                                                                 |
| Rated power                                                                               | up to 2,33 MVA                                                                                                                                                                                                                                                                  |
| Grid frequency                                                                            | 50 / 60 Hz                                                                                                                                                                                                                                                                      |
| Grid rated voltage                                                                        | from 6,6 kV up to 36 kV                                                                                                                                                                                                                                                         |
| Power factor <sup>(2)</sup> / Displacement power factor <sup>(3)</sup>                    | 1,0 / 0,8 inductive to 0,8 capacitive                                                                                                                                                                                                                                           |
| Equipment                                                                                 |                                                                                                                                                                                                                                                                                 |
| Inverter                                                                                  | EFASOLAR Inverters                                                                                                                                                                                                                                                              |
| LV/MV Transformer                                                                         | Efacec Dry type / Efacec Oil immersed                                                                                                                                                                                                                                           |
| Transformer standards                                                                     | IEC 60076 standard (Ecodesing normative optional)                                                                                                                                                                                                                               |
| MV Switchgear                                                                             | Efacec Fluofix / Efacec Normafix (up to 36 kV)                                                                                                                                                                                                                                  |
| MV Switchgear functions <sup>(4)</sup>                                                    | Star topology<br>Ring topology                                                                                                                                                                                                                                                  |
|                                                                                           | IS + CIS (fuse) / IS+DC (breaker) + Protection Relay panel<br>2x IS + CIS (fuse) / 2x IS+DC (breaker) + Protection Relay panel                                                                                                                                                  |
| Protective devices                                                                        |                                                                                                                                                                                                                                                                                 |
| DC disconnect device                                                                      | Motor-drive switch disconnecter                                                                                                                                                                                                                                                 |
| DC overvoltage protection                                                                 | Type II surge arrester                                                                                                                                                                                                                                                          |
| AC disconnect device                                                                      | Fuse / Circuit breaker                                                                                                                                                                                                                                                          |
| AC overvoltage protection                                                                 | from 75 kV to 180 kV                                                                                                                                                                                                                                                            |
| Other protective devices and protective functions included                                | Auxiliaries overvoltage protection, Ground fault monitoring, Overvoltage, Undervoltage, Overfrequency, Underfrequency, Anti-islanding, Reverse polarization, Short circuit on the output, Overtemperature, Asymmetrical current, DC unbalance strings, and others               |
| General data                                                                              |                                                                                                                                                                                                                                                                                 |
| Enclosure type                                                                            | 20ft HC Container                                                                                                                                                                                                                                                               |
| Dimensions (WxLxH)                                                                        | 2438 x 6058 x 2896 mm                                                                                                                                                                                                                                                           |
| Weight                                                                                    | ≤ 22 ton                                                                                                                                                                                                                                                                        |
| Standard temperature range <sup>(5)</sup>                                                 | -20 °C ... +60 °C / -4 °F ... +140 °F                                                                                                                                                                                                                                           |
| Max. permissible value for relative humidity (noncondensing)                              | ≤ 95%                                                                                                                                                                                                                                                                           |
| Cooling concept                                                                           | Energy efficient air forced cooling system                                                                                                                                                                                                                                      |
| Altitude for rated conditions / Maximum operating altitude above sea level <sup>(6)</sup> | 1000 m / 3000 m                                                                                                                                                                                                                                                                 |
| Protection degree                                                                         | Inverter and MV switchgear compartments: IP 54 / Transformer compartment: IP21                                                                                                                                                                                                  |
| Features                                                                                  |                                                                                                                                                                                                                                                                                 |
| Auxiliary circuit transformer <sup>(7)</sup>                                              | 230 V, 50/60 Hz                                                                                                                                                                                                                                                                 |
| Auxiliary circuit board                                                                   | included                                                                                                                                                                                                                                                                        |
| LV / MV Transformer                                                                       | Transformer oil containment                                                                                                                                                                                                                                                     |
| Communication protocols                                                                   | Modbus TCP / RTU                                                                                                                                                                                                                                                                |
| Enclosure isolation                                                                       | Walls, roof and doors isolated with thermal isolation panels (except transformer room)                                                                                                                                                                                          |
| Others                                                                                    | Earth circuit, illumination, power outlets, safety equipment, transportation parts, anti-corrosion treatment                                                                                                                                                                    |
| Optional                                                                                  |                                                                                                                                                                                                                                                                                 |
| Inverters                                                                                 | DC fused inputs configurable, Optical-Fiber communication ports                                                                                                                                                                                                                 |
| LV/MV Transformer                                                                         | Transformer alarm contacts (thermometer, DGPT or others)                                                                                                                                                                                                                        |
| MV Switchgear                                                                             | MV circuit breaker (DC function), CTs and VTs in MV modules, MV protection relay with progressive auto-recloser                                                                                                                                                                 |
| Others                                                                                    | UPS, Communications board, Air conditioning, Auxiliary transformer for trackers supply, dust protection filter levels, Meteorological Station, Efacec Monitoring systems, Efacec Management solutions, intrusion and fire detection system, external lighting system and others |

(1) Other configurations can be used.  
(2) Power factor > 0,98 at rated output voltage and power load > 15%.  
(3) The adjustable range can be extended and other values can be configured.  
(4) Other configurations can be used.  
(5) Please consult the EFASOLAR inverter technical information about the power derate with temperature within the indicated operating range.  
(6) Please consult Efacec with the specific operating conditions in order to characterize an eventual derate with altitude.  
(7) Several rated power are available up to 30 kVA.



EFASOLAR PVStation 40ft Container

The right solution for utility scale PV power plant for overseas projects. This Efacec PVStation standard container size is a plug & play solution for all kind of projects, with several cooling options, covering all environmental conditions.

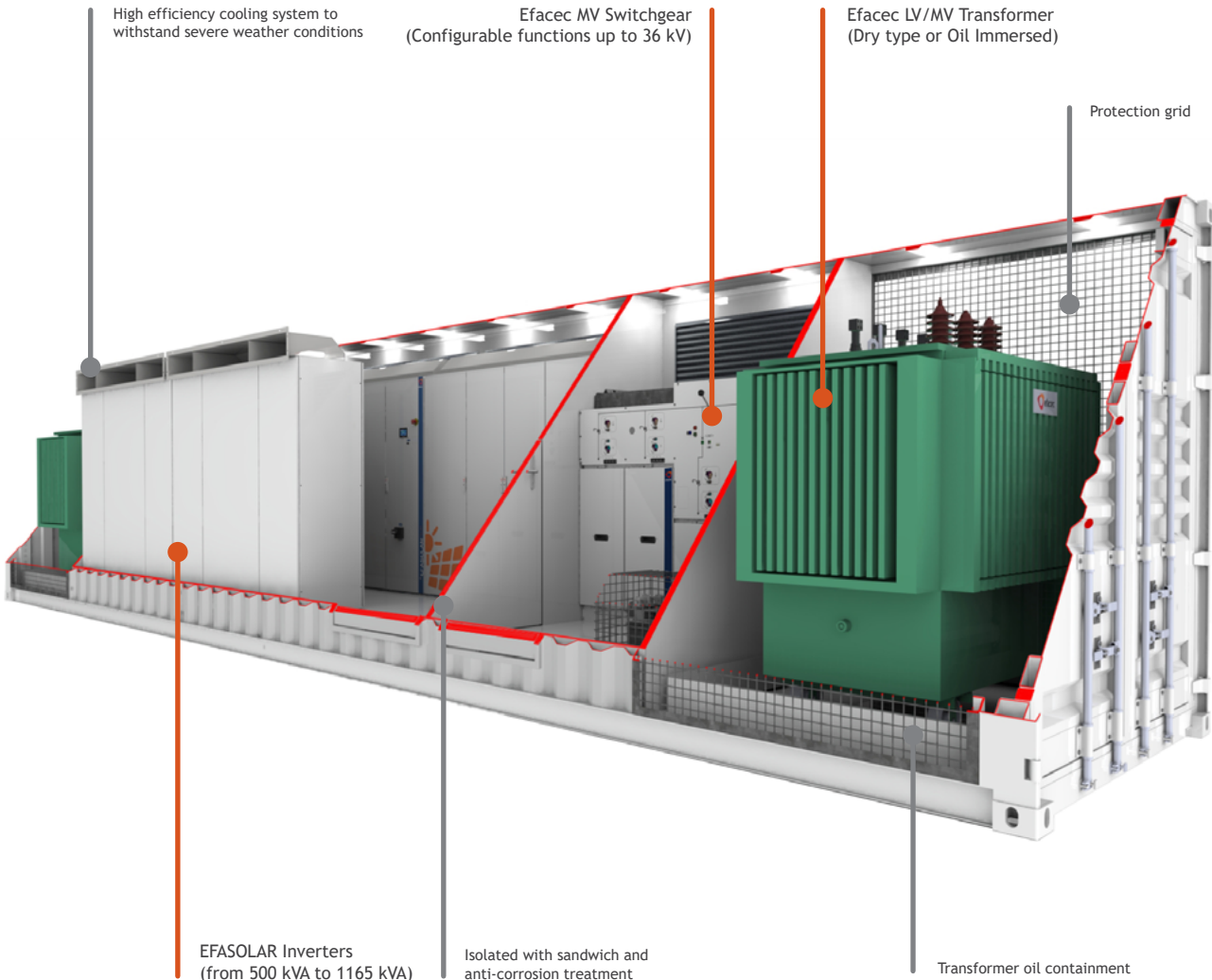


Customer Benefits

- 40ft solutions up to 4,66 MVA
- Standard transportation
- Plug & play solution
- Configurable MV switchgear functions
- High efficiency cooling system

Key Benefits

- All Efacec equipment are manufactured in-house
- Efacec single source of technology
- One single warranty for entire PVStation
- One service support for all products
- Robust and high quality container



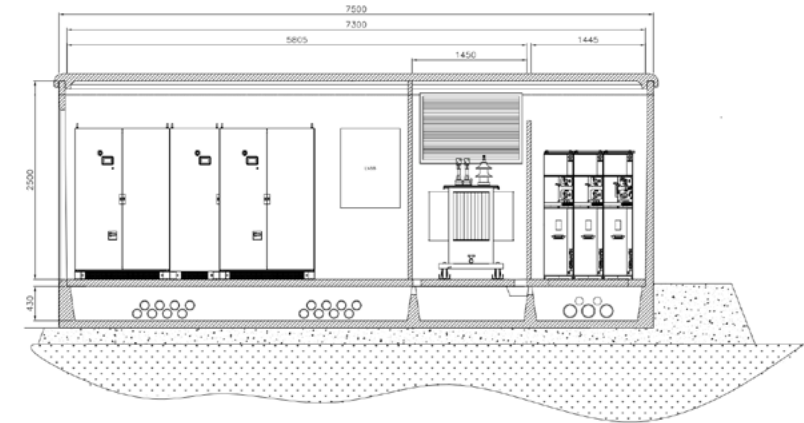
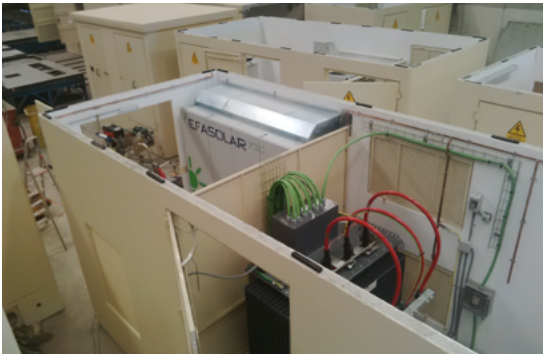
| Electrical                                                                                |                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC input                                                                                  |                                                                                                                                                                                                                                                                                 |
| Number of independent MPP inputs                                                          | 4                                                                                                                                                                                                                                                                               |
| Number of DC fused inputs <sup>(1)</sup>                                                  | up to 24                                                                                                                                                                                                                                                                        |
| AC output                                                                                 |                                                                                                                                                                                                                                                                                 |
| Rated power                                                                               | up to 4,66 MVA                                                                                                                                                                                                                                                                  |
| Grid frequency                                                                            | 50 / 60 Hz                                                                                                                                                                                                                                                                      |
| Grid rated voltage                                                                        | from 6,6 kV up to 36 kV                                                                                                                                                                                                                                                         |
| Power factor <sup>(2)</sup> / Displacement power factor <sup>(3)</sup>                    | 1,0 / 0,8 inductive to 0,8 capacitive                                                                                                                                                                                                                                           |
| Equipment                                                                                 |                                                                                                                                                                                                                                                                                 |
| Inverter                                                                                  | EFASOLAR Inverters                                                                                                                                                                                                                                                              |
| LV/MV Transformer                                                                         | Efacec Dry type / Efacec Oil immersed                                                                                                                                                                                                                                           |
| Transformer standards                                                                     | IEC 60076 standard (Ecodesing normative optional)                                                                                                                                                                                                                               |
| MV Switchgear                                                                             | Efacec Fluofix / Efacec Normafix (up to 36 kV)                                                                                                                                                                                                                                  |
| MV Switchgear functions <sup>(4)</sup>                                                    | Star topology / IS + CIS (fuse) / IS + 2x CIS (fuse) / IS+DC (breaker) / IS+ 2x DC (breaker) + Protection Relay panel                                                                                                                                                           |
|                                                                                           | Ring topology / 2x IS + CIS (fuse) / 2x IS + 2x CIS (fuse) / 2x IS+DC (breaker) / 2x IS+ 2x DC (breaker) + Protection Relay panel                                                                                                                                               |
| Protective devices                                                                        |                                                                                                                                                                                                                                                                                 |
| DC disconnect device                                                                      | Motor-drive switch disconnecter                                                                                                                                                                                                                                                 |
| DC overvoltage protection                                                                 | Type II surge arrester                                                                                                                                                                                                                                                          |
| AC disconnect device                                                                      | Fuse / Circuit breaker                                                                                                                                                                                                                                                          |
| AC overvoltage protection                                                                 | from 75 kV to 180 kV                                                                                                                                                                                                                                                            |
| Other protective devices and protective functions included                                | Auxiliaries overvoltage protection, Ground fault monitoring, Overvoltage, Undervoltage, Overfrequency, Underfrequency, Anti-islanding, Reverse polarization, Short circuit on the output, Overtemperature, Asymmetrical current, DC unbalance strings, and others               |
| General data                                                                              |                                                                                                                                                                                                                                                                                 |
| Enclosure type                                                                            | 40ft HC Container                                                                                                                                                                                                                                                               |
| Dimensions (WxLxH)                                                                        | 2438 x 12116 x 2896 mm                                                                                                                                                                                                                                                          |
| Weight                                                                                    | ≤ 28 ton                                                                                                                                                                                                                                                                        |
| Standard temperature range <sup>(5)</sup>                                                 | -20 °C ... +60 °C / -4 °F ... +140 °F                                                                                                                                                                                                                                           |
| Max. permissible value for relative humidity (noncondensing)                              | ≤ 95%                                                                                                                                                                                                                                                                           |
| Cooling concept                                                                           | Energy efficient air forced cooling system                                                                                                                                                                                                                                      |
| Altitude for rated conditions / Maximum operating altitude above sea level <sup>(6)</sup> | 1000 m / 3000 m                                                                                                                                                                                                                                                                 |
| Protection degree                                                                         | Inverter and MV switchgear compartments: IP 54 / Transformer compartment: IP21                                                                                                                                                                                                  |
| Features                                                                                  |                                                                                                                                                                                                                                                                                 |
| Auxiliary circuit transformer <sup>(7)</sup>                                              | 230 V, 50/60 Hz                                                                                                                                                                                                                                                                 |
| Auxiliary circuit board                                                                   | included                                                                                                                                                                                                                                                                        |
| LV / MV Transformer                                                                       | Transformer oil containment                                                                                                                                                                                                                                                     |
| Communication protocols                                                                   | Modbus TCP / RTU                                                                                                                                                                                                                                                                |
| Enclosure isolation                                                                       | Walls, roof and doors isolated with thermal isolation panels (except transformer room)                                                                                                                                                                                          |
| Others                                                                                    | Earth circuit, illumination, power outlets, safety equipment, transportation parts, anti-corrosion treatment.                                                                                                                                                                   |
| Optional                                                                                  |                                                                                                                                                                                                                                                                                 |
| Inverters                                                                                 | DC fused inputs configurable, Optical-Fiber communication ports                                                                                                                                                                                                                 |
| LV/MV Transformer                                                                         | Transformer alarm contacts (thermometer, DGPT or others)                                                                                                                                                                                                                        |
| MV Switchgear                                                                             | MV circuit breaker (DC function), CTs and VTs in MV modules, MV protection relay with progressive auto-recloser                                                                                                                                                                 |
| Others                                                                                    | UPS, Communications board, Air conditioning, Auxiliary transformer for trackers supply, dust protection filter levels, Meteorological Station, Efacec Monitoring systems, Efacec Management solutions, intrusion and fire detection system, external lighting system and others |

(1) Other configurations can be used.  
(2) Power factor > 0,98 at rated output voltage and power load > 15%.  
(3) The adjustable range can be extended and other values can be configured.  
(4) Other configurations can be used.  
(5) Please consult the EFASOLAR inverter technical information about the power derate with temperature within the indicated operating range.  
(6) Please consult Efacec with the specific operating conditions in order to characterize an eventual derate with altitude.  
(7) Several rated power are available up to 30 kVA.

EFASOLAR PVStation Concrete

PVStation with improved robustness with all Efacec equipment in a plug & play mode.

Full solution designed to support the most severe weather conditions to ensure the highest performance. It is a robust solution with fast installation, easy maintenance allowing to minimized the OPEX and maximize the energy generation.

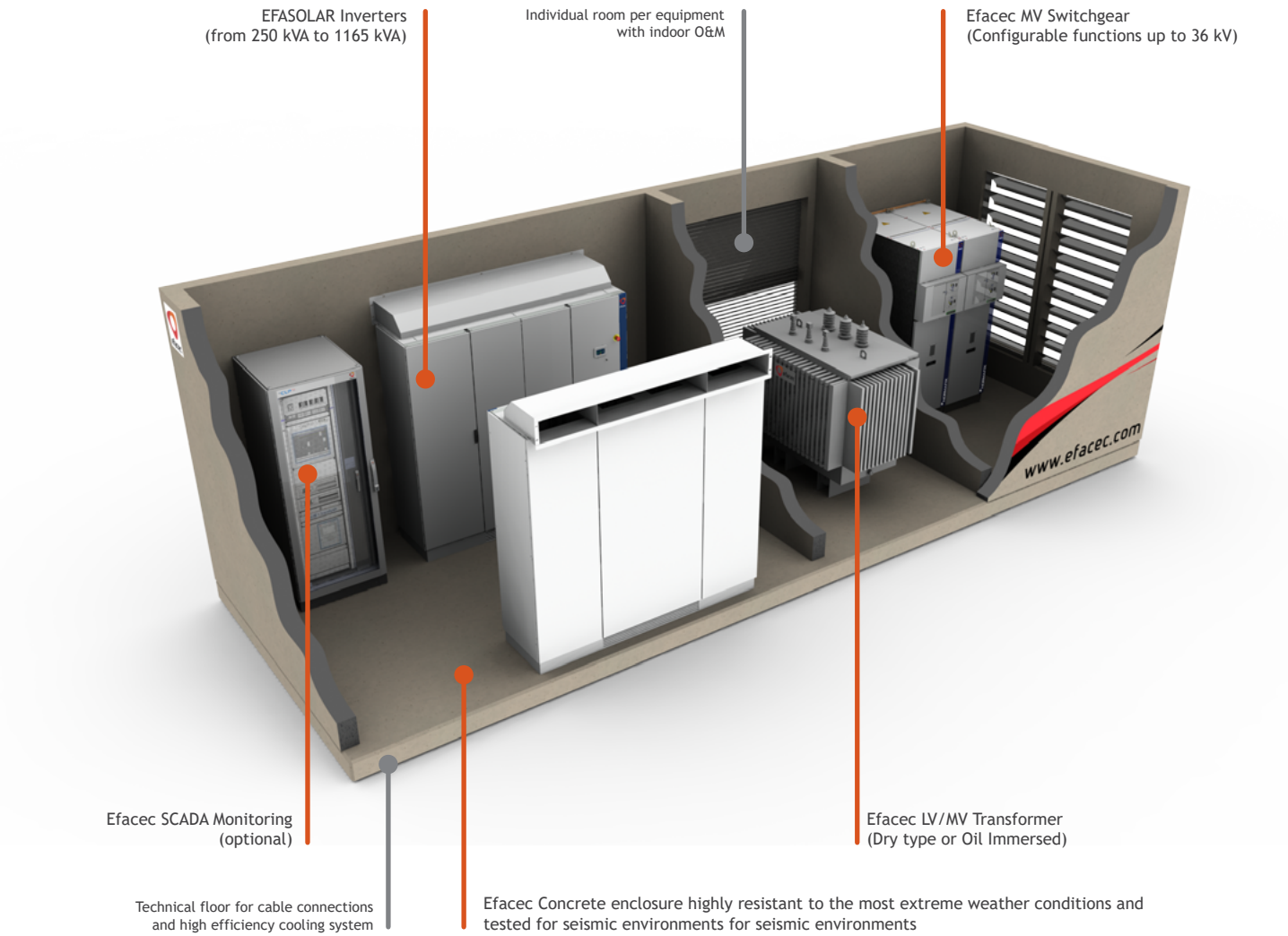


Customer Benefits

- Solutions up to 2 MVA
- Reliability and Robustness solution
- Low temperature variation inside PVStation
- Plug & play solution
- Configurable MV switchgear functions

Key Benefits

- All Efacec equipment are manufactured in-house
- Efacec single source of technology
- One single warranty for entire PVStation
- One service support for all products
- Ready to integrate more Efacec products



| Electrical                                                                                |                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC input                                                                                  |                                                                                                                                                                                                                                                                                 |
| Number of independent MPP inputs                                                          | 4                                                                                                                                                                                                                                                                               |
| Number of DC fused inputs <sup>(1)</sup>                                                  | up to 12                                                                                                                                                                                                                                                                        |
| AC output                                                                                 |                                                                                                                                                                                                                                                                                 |
| Rated power                                                                               | up to 2 MVA                                                                                                                                                                                                                                                                     |
| Grid frequency                                                                            | 50 / 60 Hz                                                                                                                                                                                                                                                                      |
| Grid rated voltage                                                                        | from 6,6 kV up to 36 kV                                                                                                                                                                                                                                                         |
| Power factor <sup>(2)</sup> / Displacement power factor <sup>(3)</sup>                    | 1,0 / 0,8 inductive to 0,8 capacitive                                                                                                                                                                                                                                           |
| Equipment                                                                                 |                                                                                                                                                                                                                                                                                 |
| Inverter                                                                                  | EFASOLAR Inverters                                                                                                                                                                                                                                                              |
| LV/MV Transformer                                                                         | Efacec Dry type / Efacec Oil immersed                                                                                                                                                                                                                                           |
| Transformer standards                                                                     | IEC 60076 standard (Ecodesign normative optional)                                                                                                                                                                                                                               |
| MV Switchgear                                                                             | Efacec Fluofix / Efacec Normafix (up to 36 kV)                                                                                                                                                                                                                                  |
| MV Switchgear functions <sup>(4)</sup>                                                    | Star topology<br>Ring topology                                                                                                                                                                                                                                                  |
|                                                                                           | IS + CIS (fuse) / IS+DC (breaker) + Protection Relay panel<br>2x IS + CIS (fuse) / 2x IS+DC (breaker) + Protection Relay panel                                                                                                                                                  |
| Protective devices                                                                        |                                                                                                                                                                                                                                                                                 |
| DC disconnect device                                                                      | Motor-drive switch disconnecter                                                                                                                                                                                                                                                 |
| DC overvoltage protection                                                                 | Type II surge arrester                                                                                                                                                                                                                                                          |
| AC disconnect device                                                                      | Fuse / Circuit breaker                                                                                                                                                                                                                                                          |
| AC overvoltage protection                                                                 | from 75 kV to 180 kV                                                                                                                                                                                                                                                            |
| Other protective devices and protective functions included                                | Auxiliaries overvoltage protection, Ground fault monitoring, Overvoltage, Undervoltage, Overfrequency, Underfrequency, Anti-islanding, Reverse polarization, Short circuit on the output, Overtemperature, Asymmetrical current, DC unbalance strings, and others               |
| General data                                                                              |                                                                                                                                                                                                                                                                                 |
| Enclosure type                                                                            | Concrete                                                                                                                                                                                                                                                                        |
| Dimensions (WxLxH)                                                                        | 4500 / 5500 / 6500 / 7500 / 8500 x 2520 x 2600 mm                                                                                                                                                                                                                               |
| Weight                                                                                    | ≤ 35 ton                                                                                                                                                                                                                                                                        |
| Standard temperature range <sup>(5)</sup>                                                 | -20 °C ... +60 °C / -4 °F ... +140 °F                                                                                                                                                                                                                                           |
| Max. permissible value for relative humidity (noncondensing)                              | ≤ 95%                                                                                                                                                                                                                                                                           |
| Cooling concept                                                                           | Energy efficient air forced cooling system                                                                                                                                                                                                                                      |
| Altitude for rated conditions / Maximum operating altitude above sea level <sup>(6)</sup> | 1000 m / 3000 m                                                                                                                                                                                                                                                                 |
| Protection degree                                                                         | Inverter, Transformer and MV switchgear compartments: IP 54                                                                                                                                                                                                                     |
| Features                                                                                  |                                                                                                                                                                                                                                                                                 |
| Auxiliary circuit transformer <sup>(7)</sup>                                              | 230 V, 50/60 Hz                                                                                                                                                                                                                                                                 |
| Auxiliary circuit board                                                                   | included                                                                                                                                                                                                                                                                        |
| LV / MV Transformer                                                                       | Transformer oil containment                                                                                                                                                                                                                                                     |
| Communication protocols                                                                   | Modbus TCP / RTU                                                                                                                                                                                                                                                                |
| Enclosure isolation                                                                       | resistance higher than 300 kg/cm² and full impermeability                                                                                                                                                                                                                       |
| Others                                                                                    | Earth circuit, illumination, power outlets, safety equipment, transportation parts, anti-corrosion treatment                                                                                                                                                                    |
| Optional                                                                                  |                                                                                                                                                                                                                                                                                 |
| Inverters                                                                                 | DC fused inputs configurable, Optical-Fiber communication ports                                                                                                                                                                                                                 |
| LV/MV Transformer                                                                         | Transformer alarm contacts (thermometer, DGPT or others)                                                                                                                                                                                                                        |
| MV Switchgear                                                                             | MV circuit breaker (DC function), CTs and VTs in MV modules, MV protection relay with progressive auto-recloser                                                                                                                                                                 |
| Others                                                                                    | UPS, Communications board, Air conditioning, Auxiliary transformer for trackers supply, dust protection filter levels, Meteorological Station, Efacec Monitoring systems, Efacec Management solutions, intrusion and fire detection system, external lighting system and others |

(1) Other configurations can be used.  
(2) Power factor > 0,98 at rated output voltage and power load > 15%.  
(3) The adjustable range can be extended and other values can be configured.  
(4) Other configurations can be used.  
(5) Please consult the EFASOLAR inverter technical information about the power derate with temperature within the indicated operating range.  
(6) Please consult Efacec with the specific operating conditions in order to characterize an eventual derate with altitude.  
(7) Several rated power are available up to 30 kVA.



EFASOLAR PVStation Modular

With concrete or metallic solution these PVStation are designed to simplify the transport operation in projects with difficult access, or correspondance with particular requirements. The local integration of the outdoor transformer and the inverters and MV switchgear enclosure(s), ensures the necessary flexibility in terms of logistics and compliance with local requirements, while maintaining the high level of performance and reliability.

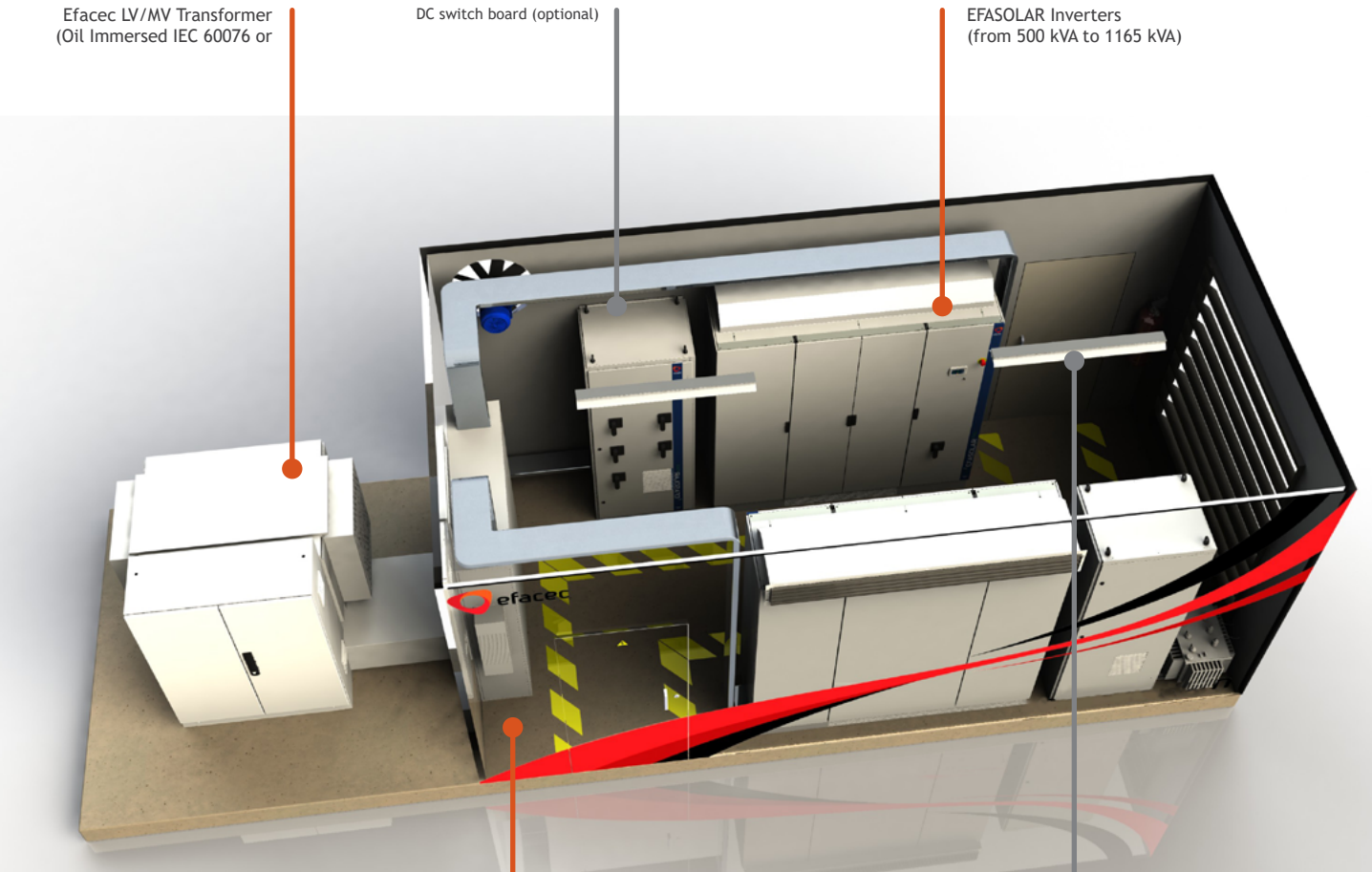


Customer Benefits

- Solutions up to 4,66 MVA
- Flexible solution
- Reliability and Robustness solution
- Customization available to each project
- Configurable MV switchgear functions

Key Benefits

- All Efacec equipment are manufactured in-house
- Efacec single source of technology
- One single warranty for entire PVStation
- One service support for all products
- Modular blocks to simplify the transportation



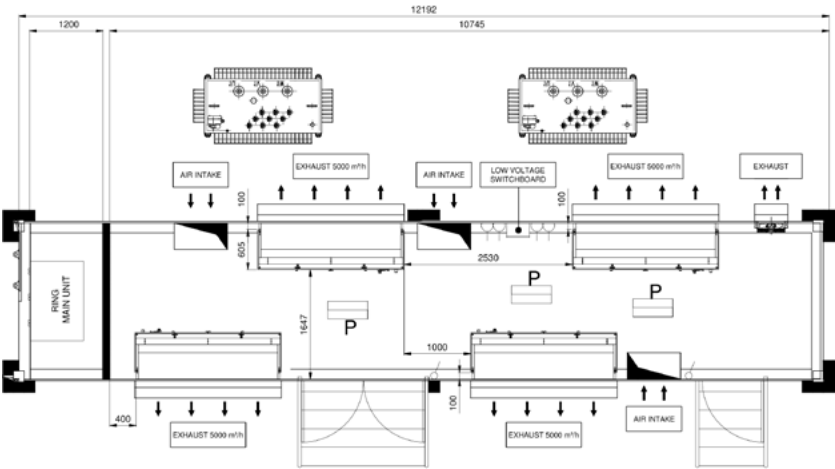
Efacec LV/MV Transformer  
(Oil Immersed IEC 60076 or

DC switch board (optional)

EFASOLAR Inverters  
(from 500 kVA to 1165 kVA)

Efacec Concrete or 20ft / 40ft Container enclosure

Illumination and other features included



| Electrical                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DC input                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                 |
| Number of independent MPP inputs                                                                                                                                                                                                                                  | 4                                                                                                                                                                                                                                                                               |
| Number of DC fused inputs <sup>(1)</sup>                                                                                                                                                                                                                          | up to 24                                                                                                                                                                                                                                                                        |
| AC output                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                 |
| Rated power                                                                                                                                                                                                                                                       | up to 4,66 MVA                                                                                                                                                                                                                                                                  |
| Grid frequency                                                                                                                                                                                                                                                    | 50 / 60 Hz                                                                                                                                                                                                                                                                      |
| Grid rated voltage                                                                                                                                                                                                                                                | from 6,6 kV up to 36 kV                                                                                                                                                                                                                                                         |
| Power factor <sup>(2)</sup> / Displacement power factor <sup>(3)</sup>                                                                                                                                                                                            | 1,0 / 0,8 inductive to 0,8 capacitive                                                                                                                                                                                                                                           |
| Equipment                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                 |
| Inverter                                                                                                                                                                                                                                                          | EFASOLAR Inverters                                                                                                                                                                                                                                                              |
| LV/MV Transformer                                                                                                                                                                                                                                                 | Efacec Oil immersed                                                                                                                                                                                                                                                             |
| Transformer standards                                                                                                                                                                                                                                             | IEC 60076 standard (Ecodesing normative optional)                                                                                                                                                                                                                               |
| MV Switchgear                                                                                                                                                                                                                                                     | Efacec Fluofix / Efacec Normafix (up to 36 kV)                                                                                                                                                                                                                                  |
| MV Switchgear functions <sup>(4)</sup>                                                                                                                                                                                                                            | Star topology<br>Ring topology                                                                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                   | IS + CIS (fuse) / IS + 2x CIS (fuse) / IS+DC (breaker) / IS+ 2x DC (breaker) + Protection Relay panel<br>2x IS + CIS (fuse) / 2x IS + 2x CIS (fuse) / 2x IS+DC (breaker) / 2x IS+ 2x DC (breaker) + Protection Relay panel                                                      |
| Protective devices                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                 |
| DC disconnect device                                                                                                                                                                                                                                              | Motor-drive switch disconnecter                                                                                                                                                                                                                                                 |
| DC overvoltage protection                                                                                                                                                                                                                                         | Type II surge arrester                                                                                                                                                                                                                                                          |
| AC disconnect device                                                                                                                                                                                                                                              | Fuse / Circuit breaker                                                                                                                                                                                                                                                          |
| AC overvoltage protection                                                                                                                                                                                                                                         | from 75 kV to 180 kV                                                                                                                                                                                                                                                            |
| Other protective devices and protective functions included                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                 |
| Auxiliaries overvoltage protection, Ground fault monitoring, Overvoltage, Undervoltage, Overfrequency, Underfrequency, Anti-islanding, Reverse polarization, Short circuit on the output, Overtemperature, Asymmetrical current, DC unbalance strings, and others |                                                                                                                                                                                                                                                                                 |
| General data                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                 |
| Enclosure type                                                                                                                                                                                                                                                    | Concrete or 20ft / 40ft Container                                                                                                                                                                                                                                               |
| Dimensions (WxLxH)                                                                                                                                                                                                                                                | 4500 / 5500 / 6500 / 7500 / 8500 x 2520 x 2600 mm or 20ft / 40ft container size                                                                                                                                                                                                 |
| Weight                                                                                                                                                                                                                                                            | ≤ 35 ton                                                                                                                                                                                                                                                                        |
| Standard temperature range <sup>(5)</sup>                                                                                                                                                                                                                         | -20 °C ... +60 °C / -4 °F ... +140 °F                                                                                                                                                                                                                                           |
| Max. permissible value for relative humidity (noncondensing)                                                                                                                                                                                                      | ≤ 95%                                                                                                                                                                                                                                                                           |
| Cooling concept                                                                                                                                                                                                                                                   | Energy efficient air forced cooling system                                                                                                                                                                                                                                      |
| Altitude for rated conditions / Maximum operating altitude above sea level <sup>(6)</sup>                                                                                                                                                                         | 1000 m / 3000 m                                                                                                                                                                                                                                                                 |
| Protection degree                                                                                                                                                                                                                                                 | Inverter and MV switchgear compartments: IP 54 / Transformer: IP65                                                                                                                                                                                                              |
| Features                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                 |
| Auxiliary circuit transformer <sup>(7)</sup>                                                                                                                                                                                                                      | 230 V, 50/60 Hz                                                                                                                                                                                                                                                                 |
| Auxiliary circuit board                                                                                                                                                                                                                                           | included                                                                                                                                                                                                                                                                        |
| LV / MV Transformer                                                                                                                                                                                                                                               | Transformer oil containment                                                                                                                                                                                                                                                     |
| Communication protocols                                                                                                                                                                                                                                           | Modbus TCP / RTU                                                                                                                                                                                                                                                                |
| Enclosure isolation                                                                                                                                                                                                                                               | Concrete<br>20ft/40ft Container                                                                                                                                                                                                                                                 |
|                                                                                                                                                                                                                                                                   | resistance higher than 300 kg/cm² and full impermeability<br>Walls, roof and doors isolated with thermal isolation panels (except transformer room)                                                                                                                             |
| Others                                                                                                                                                                                                                                                            | Earth circuit, illumination, power outlets, safety equipment, transportation parts, anti-corrosion treatment                                                                                                                                                                    |
| Optional                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                 |
| Inverters                                                                                                                                                                                                                                                         | DC fused inputs configurable, Optical-Fiber communication ports                                                                                                                                                                                                                 |
| LV/MV Transformer                                                                                                                                                                                                                                                 | Transformer alarm contacts (thermometer, DGPT or others)                                                                                                                                                                                                                        |
| MV Switchgear                                                                                                                                                                                                                                                     | MV circuit breaker (DC function), CTs and VTs in MV modules, MV protection relay with progressive auto-recloser                                                                                                                                                                 |
| Others                                                                                                                                                                                                                                                            | UPS, Communications board, Air conditioning, Auxiliary transformer for trackers supply, dust protection filter levels, Meteorological Station, Efacec Monitoring systems, Efacec Management solutions, intrusion and fire detection system, external lighting system and others |

(1) Other configurations can be used.  
(2) Power factor > 0,98 at rated output voltage and power load > 15%.  
(3) The adjustable range can be extended and other values can be configured.  
(4) Other configurations can be used.  
(5) Please consult the EFASOLAR inverter technical information about the power derate with temperature within the indicated operating range.  
(6) Please consult Efacec with the specific operating conditions in order to characterize an eventual derate with altitude.  
(7) Several rated power are available up to 30 kVA.



**Efacec Energia, Máquinas e Equipamentos Eléctricos, S.A.**

High and Medium Voltage Switchgear

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